Excercise Machine Learning -2

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1. Look at table 1. Determine the class (Y) for the test data using the 3-nearest neighbor method. Please use the euclidian distance for the distance calculation process!

Table 1. Training and Testing data

	Training Data						Testing Data	
X1	2	1	2	3	1	3	2	1
X2	2	2	3	3	0	2	1	1
Y	-1	1	-1	1	1	-1	???	???

2. Please explain for which data the Support Vector Machine is suitable!

3. Please pay attention to Figure 1. Please explain how the machine learning method works in Figure 1.

```
from sklearn import naive_bayes

nb = naive_bayes.CategoricalNB()

nb.fit(X_train, y_train)
y_pred = nb.predict(X_test)
```

Figure 1. Code snippets for number 3

4. a. Please explain the algorithms of bagging, boosting and stacking! Please also explain the diffencess! (score : 3%)

   b. Look at Figure 2. Please explain what is the meaning of base classifier, n_estimators and learning rate! (score:2%)

```
    adaboost_classifier = AdaBoostClassifier(
        base_classifier, n_estimators=50, learning_rate=1.0, random_state=42
    )
adaboost_classifier.fit(X_train, y_train)
```

Figure 2. Code Snippets of Adaboost

   c. Please pay attention to the following data. Please implement the Bagging algorithm for the data. Please use only 3 bootstrapings/single models! Please write the final prediction and calculate the accuracy! (score : 5%)

Table 2. Data for Bagging Algorithm

Source Dataset	x	2,4	3,1	3,7	4,4	5,2	5,8
	y	1	1	1	-1	-1	-1